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ECB Communications as Monetary Shocks

A Multi-Agent LLM Framework and Asymmetric Transmission Analysis

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Abstract

This thesis develops a multi-agent large language model (LLM) pipeline to extract monetary policy surprise measures from the official communications of the European Central Bank (ECB) over the period March 1999 to December 2025, covering all 312 Governing Council meetings. A four-stage pipeline — two large language model agents followed by two deterministic aggregation modules — processes documents publicly available before each meeting's blackout period (Accounts of previous meetings, Economic Bulletins, and official speeches) and generates a probability distribution over possible rate decisions. The document inputs are *ex-ante* by construction and the measure is *market neutral*, being built exclusively from institutional narrative without incorporating any asset price information. A separate limitation — distinct from market neutrality — is that the language model reading these documents was trained on data extending beyond the sample period, potentially introducing temporal bias through both interpretive and outcome-leakage channels; this is discussed in detail in Chapter 3.

We document three empirical regularities in the surprise series: surprises are on average close to zero across the full sample, consistent with broadly informative ECB communications; surprise volatility is lowest during the Draghi presidency, reflecting a combination of explicit forward guidance and policy rates pinned near the effective lower bound; and the largest surprises in the sample correspond to the unprecedented 75 bp hikes of September and October 2022, where official communications materially understated the forthcoming decisions.

In a second stage, we use the surprise series as a monetary policy shock in a Local Projections framework (Jordà, 2005) to estimate the dynamic transmission of ECB surprises on euro area HICP inflation — with GDP growth included as a conditioning control and separately examined as a dependent variable in a supplementary check. The main empirical finding is a statistically significant asymmetry: hawkish surprises are associated with a reduction in HICP inflation of approximately 0.86 percentage points nine months after the shock, while dovish surprises of equal magnitude produce an effect approximately 56 % as large — a ratio of roughly 1.8 to 1. This asymmetry is robust across fourteen out of twenty-four horizons, confirmed by Wald tests for equality of coefficients.

Keywords: Monetary policy surprises, European Central Bank, Large Language Models, Local Projections, Natural language processing, Asymmetric transmission.

JEL Classification: E52, E58, C55, C32, G14.

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1

Introduction

The way central banks communicate has changed profoundly over the past three decades. What was once considered a virtue — the deliberate opacity of monetary policy, epitomised by Alan Greenspan's famous remark that "if I seem unduly clear to you, you must have misunderstood what I said" — has given way to an era of radical transparency. Central banks now publish detailed accounts of their internal deliberations, deliver structured press conferences, and guide markets with explicit forward guidance months in advance. This transformation raises a natural question: if central banks communicate so much, do markets still get surprised? The evidence presented in this thesis suggests that they do — and that the pattern of those surprises contains important information about the transmission of monetary policy.

Measuring monetary policy surprises is not straightforward. The dominant approach in the literature uses high-frequency changes in interest rate derivatives — OIS rates, fed funds futures, Euribor contracts — around policy announcements as a proxy for the unexpected component of the decision. This approach has produced influential results, but it conflates two distinct phenomena: the surprise *relative to what the central bank had communicated*, and the surprise *relative to what financial markets had already priced in* through informal channels, rumours, and speeches in the days preceding the meeting. When a central bank raises rates by 25 bp and the OIS rate does not move, it is tempting to conclude there was no surprise. But if the central bank's own official communications had pointed to a hold, the decision was surprising — relative to the institutional signal, if not to the market price. This thesis takes that distinction seriously. Rather than measuring surprises from asset prices, we extract them directly from the central bank's official communications using a **multi-agent system of large language models (LLMs)**. For each of the 312 Governing Council meetings of the European Central Bank between March 1999 and December 2025, a four-stage pipeline — two LLM agents that read documents (Agents IM and IB) followed by two deterministic modules that

synthesise their outputs (Agents II and III) — processes the material that was publicly available before the meeting — Accounts of the previous meeting, Economic Bulletins, and official speeches — and produces a probability distribution over possible rate decisions. Two constraints govern the entire system. The first is **ex-ante construction**: every document used by the agents must have been published before the ten-day blackout period preceding each meeting, ensuring that no information unavailable to market participants at the time enters the expectation formation process. (The reader model itself was trained on more recent data, a qualification we return to as a limitation in Chapter 6.) The second is **market neutrality**: expectations are formed exclusively from official institutional communications, without incorporating any market price data, so that the resulting surprise measure is exogenous with respect to asset prices.

The ECB is a natural laboratory for this analysis. Unlike the Federal Reserve, it operates under a single mandate of price stability, which imposes a clear hierarchy on the information used to form expectations: inflation dominates. Its communication architecture evolved substantially over the sample period — the Accounts of the Governing Council meetings were introduced only in 2015, forward guidance became an explicit policy tool under Draghi, and the 2022–2023 hiking cycle tested the limits of the ECB's communication framework in ways that had no historical precedent — providing precisely the variation needed to identify how surprises are formed and transmitted across different institutional environments. The analysis proceeds in two stages. In the first, we document the properties of the surprise series: surprises are on average close to zero across the full sample, consistent with broadly informative ECB communications; the Draghi era exhibits the lowest surprise volatility of the three presidential periods, reflecting both explicit forward guidance and policy rates pinned near the effective lower bound; and the largest surprises concentrate in the 2022–2023 hiking cycle, when official communications repeatedly understated the pace of tightening. In the second stage, we use the surprise series as a monetary policy shock in a Local Projections framework (Jordà, 2005) to estimate its dynamic effects on euro area HICP inflation. The main finding is a statistically significant asymmetry: hawkish surprises are associated with a reduction in inflation of approximately 0.86 percentage points nine months after the shock, while dovish surprises of equal magnitude predict an effect approximately 56 % as large — a ratio of roughly 1.8 to 1. This asymmetry holds across fourteen out of twenty-four horizons and is confirmed by Wald tests for equality of coefficients.

This thesis pursues three connected objectives: constructing a multi-agent LLM pipeline adapted to the ECB institutional context, producing a narrative surprise series covering the full 1999–2025 period, and testing for asymmetric transmission

using Local Projections. Each objective builds on existing work — in particular the framework of Fernández-Fuertes (2025) — and the results are intended as a contribution to an ongoing research conversation rather than a definitive settlement of open questions. The remainder of the thesis is organised as follows. Chapter 2 reviews the related literature on monetary policy surprise measurement, central bank communication, and asymmetric transmission. Chapter 3 describes the data sources and pipeline architecture. Chapter 4 presents the descriptive properties of the surprise dataset. Chapter 5 reports the econometric results. Chapter 6 concludes.

2

Literature Review

This chapter reviews the three strands of literature on which this thesis builds: the measurement of monetary policy surprises, the application of natural language processing to central bank communications, and the estimation of asymmetric monetary transmission using Local Projections.

2.1 Measuring Monetary Policy Surprises

The systematic measurement of monetary policy surprises begins with Kuttner (2001), who demonstrated that changes in the federal funds futures rate around FOMC announcements provide a clean decomposition of the policy decision into its anticipated and unanticipated components. The intuition is straightforward: financial markets continuously update their expectations about future policy, and the change in futures prices at the moment of the announcement captures only the component that was not already priced in. Kuttner (2001) showed that this unanticipated component generates significant responses in Treasury yields, while the anticipated component does not — a finding that validated the identification strategy and motivated a large subsequent literature.

Gürkaynak et al. (2005) extended this framework in two directions that are directly relevant to this thesis. First, they documented that monetary policy announcements move asset prices along two dimensions: a pure rate surprise and a path factor capturing revisions to the expected future trajectory of rates. This two-factor structure implies that central bank communication about future intentions — forward guidance — has independent effects on financial markets beyond the immediate rate decision, a finding that motivates our focus on the full probability distribution over rate outcomes rather than a scalar surprise measure. Second, they showed that the path factor is driven primarily by the content of the post-meeting statement rather than the rate decision itself, providing early evidence that the informational content of central bank communications extends well beyond the announced decision.

The identification approach underlying both papers rests on an assumption that Nakamura and Steinsson (2018) called into question: that the only information released at a policy announcement is the rate decision. Using a structural approach that jointly identifies monetary policy shocks and central bank information shocks, they document that FOMC announcements frequently contain news about the economic outlook — when the Federal Reserve raises rates and stock prices rise simultaneously, markets are inferring positive information about growth prospects, not tightening financial conditions. Failing to account for this information effect generates the well-documented price puzzle: the apparently expansionary short-run response of output and inflation to monetary tightening. Jarociński and Karadi (2020) apply a related identification to the ECB, finding that nearly half of all Governing Council announcements are accompanied by a positive co-movement of interest rates and equity prices, consistent with a large and systematic information component in ECB communications. The narrative surprise measure constructed in this thesis is designed to sidestep this problem: by measuring the gap between official communications and decisions rather than the asset price response, we construct a narrative policy-decision gap that is not mechanically contaminated by announcement-window asset-price responses.

2.2 Central Bank Communication and Text Analysis

The recognition that central bank communication is itself a policy instrument predates the NLP literature. Blinder et al. (2008) provided a systematic treatment of the economics of central bank communication, arguing that transparency and predictability reduce the costs of monetary policy implementation by aligning private sector expectations with the central bank's own outlook. The question of how to measure the informational content of central bank texts, however, required new methodological tools.

Early quantitative work relied on manual coding of policy statements. Ehrmann and Fratzscher (2007) studied how the heterogeneity of communications across ECB Governing Council members affects financial market outcomes, documenting that the ECB President's speeches carry systematically larger market impact than those of other members — a finding that directly informs our speaker-weighting scheme in Agent IB. Altavilla et al. (2019) construct the Euro Area Monetary Policy Event-Study Database (EA-MPD), mapping ECB policy communication into yield curve changes and identifying three distinct factors — target rate, forward guidance, and quantitative easing — that capture different segments of the yield curve. Their high-frequency dataset provides the market-based benchmark against which our narrative measure can be compared.

The application of machine learning methods transformed this literature. Hansen et al. (2018) used computational linguistics to analyse FOMC transcripts, documenting that the introduction of transparency reforms changed the deliberative dynamics of the Committee, with members becoming more strategically minded in their communications once transcripts were published with a lag. This finding is relevant for our institutional context: the ECB's introduction of the Accounts in 2015 represents a comparable transparency reform, and the structural break we document in the surprise series at that date may partly reflect a genuine change in the information content of ECB communications rather than a pure document-substitution effect.

Large language models represent the methodological frontier of this research agenda. Unlike dictionary-based or topic model approaches, LLMs can process contextual nuance, resolve ambiguity, and extract structured information from complex institutional documents without requiring domain-specific lexicons or pre-specified categories. The closest antecedent to this thesis is Fernández-Fuertes (2025), who constructs a multi-agent LLM system to extract monetary policy surprises from Federal Reserve communications, demonstrating that the resulting narrative measure contains predictive information about future macroeconomic outcomes beyond what market-based surprise measures capture. Our work adapts this framework to the ECB institutional context, which requires a set of substantive modifications — the three-rate system, the absence of Accounts before 2015, and the single price stability mandate — described in detail in Chapter 3.

A related study by Bottazzi et al. (2026), prepared for the European Parliament's Economic Governance and EMU Scrutiny Unit, applies a multi-agent LLM framework directly to ECB communications to assess the ECB's monetary policy stance and macroeconomic uncertainty. That work shares the institutional focus of the present thesis; our contribution is complementary in scope: we cover the full 1999–2025 sample, construct a quantitative surprise series at meeting frequency, and test explicitly for direction-dependent transmission asymmetry using Local Projections, which is not the primary focus of Bottazzi et al. (2026).

A parallel strand of LLM applications to financial texts focuses not on structured scalar extraction but on high-dimensional semantic embeddings. Chen et al. (2025) generate contextualised text representations of financial news articles using large language models and use the resulting embedding vectors as predictors of cross-sectional equity returns. Their approach and ours share the premise that LLMs can extract economically meaningful content from unstructured text, but differ fundamentally in methodology and objective: Chen et al. (2025) use embeddings as a black-box predictor of asset price movements, while this thesis extracts interpretable scalar scores from institutional documents to construct a measure of

the gap between central bank communications and decisions. The interpretability of our approach is a deliberate design choice, not a limitation: it allows direct comparison with theory-based predictions and enables the identification strategy underlying the Local Projections analysis.

2.3 Local Projections and Monetary Transmission

The econometric framework for estimating the dynamic effects of monetary policy shocks has been shaped by two approaches: the structural VAR of Christiano et al. (1999) and the Local Projections method of Jordà (2005). Structural VARs impose a parametric model of the economy's dynamics and identify monetary policy shocks through a combination of short-run zero restrictions and timing assumptions. Local Projections estimate the impulse response function directly, running a sequence of regressions of the outcome variable at each horizon h on the current shock and a set of controls, without imposing a model of the intervening dynamics. Plagborg-Møller and Wolf (2021) demonstrate that Local Projections and VARs estimate the same impulse responses under unrestricted lag structures, establishing that the two approaches are not conceptually separate procedures but rather different finite-sample estimators of a common estimand — with Local Projections trading lower bias for higher variance, making them a natural framework for the asymmetry test that follows.

Romer and Romer (2004) constructed a narrative measure of Federal Reserve monetary policy shocks using internal FOMC documents to identify the component of policy changes not driven by the Fed's own assessment of the economic situation. This approach is conceptually aligned with our own: both seek to measure the unexpected component of monetary policy relative to an information set that the central bank itself possessed, rather than relative to asset prices. The Romer-Romer measure predates the LLM era and relies on manual extraction of information from structured documents; our contribution is to extend this identification approach to the ECB and to automate the information extraction through a multi-agent pipeline.

2.4 Asymmetric Monetary Transmission

The question of whether monetary tightening and easing have symmetric effects on the economy has attracted theoretical and empirical attention since at least Cover (1992), who provided early econometric evidence that negative money supply shocks have larger output effects than positive ones in the United States. The theoretical mechanisms proposed in the subsequent literature include downward nominal wage and price rigidity — which limits the expansionary effect of easing relative to the contractionary effect of tightening — and financial accelerator

dynamics, whereby tightening raises borrowing costs and tightens collateral constraints in ways that cannot be fully undone by equivalent easing.

Tenreyro and Thwaites (2016) provide relevant antecedents to our empirical results, documenting significant asymmetry in US monetary transmission using a semiparametric Local Projections approach. Their main finding concerns *state-dependent* asymmetry: monetary policy is substantially less powerful in recessions than in expansions. While conceptually distinct from the *direction-dependent* asymmetry we test — whether tightening surprises have larger effects than easing surprises of equal magnitude — both types of asymmetry share the same theoretical roots in downward nominal rigidities and financial accelerator mechanisms. Our results extend the direction-dependent evidence of Cover (1992) to the euro area using a narrative identification strategy and focusing on the inflation rather than the output channel.

For the euro area, the evidence on asymmetric transmission is more limited, partly because of the shorter history of the ECB as an independent monetary authority and partly because existing market-based surprise measures do not cleanly separate the direction of the shock from its magnitude. Our narrative measure, constructed to be symmetric around zero by design, allows a direct test of this asymmetry using a sample that spans three qualitatively distinct monetary regimes and includes both the zero lower bound period and the unprecedented tightening cycle of 2022–2023.

2.5 Positioning

Table 2.1 positions this thesis relative to the most closely related work. The distinctive features of our approach are the combination of narrative identification with Local Projections, the focus on the ECB over a long sample, and the explicit test for asymmetric transmission.

Table 2.1. *Positioning relative to related literature*

Paper	CB	Method	Period	Asymmetry	LLM
Kuttner (2001)	Fed	Futures	1989–2000	No	No
Gürkaynak et al. (2005)	Fed	Futures	1990–2004	No	No
Romer & Romer (2004)	Fed	Narrative	1969–1996	No	No
Jarociński & Karadi (2020)	Both	HF sign	1999–2016	No	No
Altavilla et al. (2019)	ECB	HF	1999–2017	No	No
Tenreyro & Thwaites (2016)	Fed	LP	1969–2007	Yes	No
Chen et al. (2025)	—	LLM emb.	2001–2021	No	Yes
Fernández-Fuertes (2025)	Fed	LLM	2000–2024	No	Yes
This thesis	**ECB**	**LLM+LP**	**1999–2025**	**Yes**	**Yes**

CB = central bank; HF = high-frequency identification; LP = Local Projections; LLM = large language model scalar extraction; LLM emb. = large language model embedding-based approach.

3

Data and Pipeline Architecture

3.1 Overview

This chapter describes the data collection process and the multi-agent LLM pipeline developed to extract monetary policy surprise measures from ECB official communications. The system is designed around two fundamental methodological constraints. The first is **ex-ante construction**: each agent may only use documents published before the ten-day pre-meeting blackout period, ensuring that no information unavailable at the time enters the formation of expectations. This applies to the document inputs; the language model that reads them was trained on more recent data, a residual qualification we treat as a limitation in Section 6.2. The second is **market neutrality**: expectations are formed exclusively from central bank communications, without incorporating market price information such as OIS rates or interest rate futures.

Market neutrality ensures that the resulting surprise measure is exogenous with respect to asset prices. Standard market-based measures — such as the change in OIS rates around the policy announcement — conflate the central bank's informational contribution with pre-existing market expectations, risk premia, and global financial conditions. By constructing expectations from official institutional sources alone, we obtain a measure that captures the gap between what the ECB communicated and what it ultimately decided, independently of how financial markets had positioned themselves. This exogeneity is with respect to asset prices, not with respect to the underlying macroeconomic information set: both the ECB and market participants observe the same latent macroeconomic variables, so the model's residuals may still be correlated with market surprises through common macroeconomic factors.

The pipeline builds on the multi-agent framework of Fernández-Fuertes (2025), originally developed for the Federal Reserve, but the ECB context imposes a set of institutional constraints that require substantive departures from the original design. The most immediate is the ECB's three-rate system. Unlike the Federal

Reserve, which sets a single policy rate, the ECB simultaneously determines the Main Refinancing Operations rate (MRO), the Deposit Facility Rate (DFR), and the Marginal Lending Facility rate (MLF). Our pipeline computes independent surprise measures for each of the three rates, preserving the full signalling content of changes in the rate corridor — a dimension that a single-rate approach would simply discard.

A more fundamental challenge concerns document availability. The Federal Reserve has published detailed meeting minutes continuously since 1993. The ECB only introduced its equivalent — the Accounts of the Monetary Policy Meeting — in January 2015. For the preceding sixteen years, which represent the majority of our sample, no comparable document of internal deliberation existed. This is not a minor data gap: the Accounts are the primary input to our monetary intelligence agent, and their absence before 2015 requires a carefully validated substitution strategy whose consequences we test formally in Section 3.6.

The ECB's institutional mandate also shapes the pipeline's design in ways that go beyond document selection. Operating under a single mandate of price stability — rather than the Federal Reserve's dual mandate — the ECB weights inflation signals differently and communicates about the labour market and output primarily through their implications for future prices. This hierarchy is embedded in the economic conditions agent through an explicit weighting structure that assigns primacy to HICP inflation assessments, calibrated to the ECB's own definition of price stability.

Finally, the temporal scope of the analysis — 312 Governing Council meetings from March 1999 to December 2025 — introduces a degree of structural variation that has no equivalent in shorter-sample studies. The sample spans the ECB's early years of credibility-building under Duisenberg, the financial and sovereign debt crises under Trichet, the forward guidance revolution and negative rates under Draghi, and the inflation surge and unprecedented tightening cycle under Lagarde. Each of these periods represents a qualitatively different communication environment, and the pipeline must produce a surprise measure that is comparable across all of them — a requirement that motivates the formal break tests and regime-interaction analyses reported in Sections 3.6 and 3.6.

3.2 Institutional Context and Document Mapping

MEETING ACCOUNTS AND PRESS CONFERENCES. The ECB publishes its Accounts of the Monetary Policy Meeting approximately four weeks after each Governing Council meeting. These documents, introduced in January 2015, provide a detailed narrative of the internal debate among Council members, including the distribution of views on the economic outlook and the appropriate policy stance.

Prior to 2015, no equivalent document existed. We use the *Introductory Statement* of the Press Conference of the previous meeting as a substitute — specifically, the portion of the transcript preceding the Q&A session, which begins with the standardised phrase "We are now at your disposal for questions." The Introductory Statement is a structured, collectively prepared document that contains explicit policy signals and forward guidance, and has been shown to be the primary driver of market reactions to ECB communications (Ehrmann and Fratzscher, 2007; Gürkaynak et al., 2005). The implications of this document substitution for the statistical properties of the surprise series are assessed formally in Section 3.6.

ECONOMIC BULLETINS. The ECB's Economic Bulletin is published eight days after each policy meeting and is therefore not available ex-ante. We use the Economic Bulletin published after the *previous* meeting — the most recent issue available before the blackout window. For the period 1999–2014, we use the *Monthly Bulletin* as its predecessor.

GOVERNING COUNCIL SPEECHES. Unlike the Federal Reserve's Beige Book, the ECB has no direct equivalent synthetic regional assessment. We address this by constructing an economic signal from official speeches delivered by Governing Council members in the 28-day window preceding each meeting, subject to the ten-day blackout constraint. Speeches are weighted by the speaker's institutional role, following the evidence in Ehrmann and Fratzscher (2007) that the ECB President's communications have systematically larger effects on financial markets than those of other Council members. Specifically, we assign weights of 1.00 to the President, 0.85 to the Vice President, 0.80 to the Chief Economist, 0.70 to Executive Board members, and 0.55 to NCB Governors. These weights are assigned *a priori* and are subject to sensitivity analysis in Appendix A.

The complete document mapping across sub-periods is summarised in Table 3.1.

3.3 Dataset Construction

3.3.1 Policy Decisions

The policy decision dataset covers all 312 Governing Council meetings from March 1999 to December 2025. For each meeting, we record the levels of the three key ECB interest rates — the MRO, the DFR, and the MLF — and the change in basis points relative to the previous meeting.

A critical implementation detail concerns the attribution of rate changes. ECB rate decisions are announced at the meeting but take effect several days later, and the standard data convention records the rate applicable at the end of the period following the meeting. Since we need to attribute the rate change to the meeting at

Table 3.1. *Document mapping across sub-periods*

Document	Period	Agent	Source
Accounts of the Monetary Policy Meeting	2015–2025	Agent IM	ECB website
Press Conference — Introductory Statement only	1999–2014	Agent IM	ECB website
Economic Bulletin (previous issue)	2015–2025	Agent IB	ECB website
Monthly Bulletin	1999–2014	Agent IB	ECB website
Governing Council speeches	1999–2025	Agent IB	ECB speech database
Monetary Policy Statement (policy decision)	1999–2025	Agent III	ECB website

IM = Monetary Intelligence; IB = Economic Conditions; III = Surprise Calculation. Agent III uses the Monetary Policy Statement exclusively to extract the announced rate decision, not as a text input to the LLM.

which it was announced, we apply a forward difference correction, shifting each rate change to the announcement meeting. This correction affects 11 observations in the sample. The DFR is our primary policy rate throughout, consistent with the literature on ECB monetary policy post-2014, when the deposit facility rate became the binding constraint on short-term money market rates. For the earlier part of the sample the MRO rate was the operationally central instrument, so the choice of the DFR as the single reference rate is a simplification; the pipeline computes surprises for all three rates, and the MRO-based mechanical surprise is highly correlated with the DFR-based measure used here, though we do not report a separate MRO re-estimation of the Local Projections.

3.3.2 ECB Communications

The communications dataset draws on five distinct document types, each contributing a different dimension of information to the pipeline.

The **Accounts** and **Press Conference Introductory Statements** are the primary input to Agent IM. The 90 Accounts PDFs collected for 2015–2025 contain the verbatim record of the internal debate within the Governing Council — the distribution of views on the economic outlook, the arguments made for and against different policy options, and the precise wording of the forward guidance agreed upon. We collect 267 Press Conference HTML transcripts spanning the full 1999–2026 period,

of which 178 predate January 2015 and serve as the primary input to Agent IM for the pre-2015 subsample. The Introductory Statement is extracted programmatically by truncating the transcript at the phrase "We are now at your disposal for questions."

A note on document coverage reconciles the input counts with the 312 meetings. The ECB held bi-weekly Governing Council sessions from March 1999 through December 2001 — roughly 68 sessions in three years — but held press conferences only monthly, typically at the first meeting of each month. The approximately 35 bi-weekly meetings that fell between monthly press conferences therefore share the Introductory Statement of the most recent monthly conference with the preceding meeting; Agent IM uses the same document twice in those cases. The remaining gap between 267 press conference transcripts and 312 total meetings ($312 - 267 = 45$) includes those ~ 35 bi-weekly sessions plus a small number of additional non-monetary-policy Governing Council sessions and unscheduled emergency meetings (such as the coordinated inter-meeting cut of October 8, 2008) for which no separate press conference transcript was published. The minor discrepancy between the 90 Accounts PDFs collected and the 87 post-2015 meeting records in the dataset reflects a small number of supplementary or revised Account documents included in the archive; it does not affect the analysis since the post-2015 subsample is defined by meeting date rather than by document count.

The **Economic Bulletins** and **Monthly Bulletins** are the primary input to Agent IB. Published in the weeks following each meeting, these documents provide the ECB's most comprehensive and structured assessment of the economic outlook — covering inflation, growth, labour markets, monetary aggregates, and financial conditions. We collect 87 Economic Bulletin PDFs for 2015–2025 and 192 Monthly Bulletin PDFs for 1999–2014, using the issue published after the previous meeting as the most recent available before the blackout window.

The **speeches** dataset, drawn from the ECB's official speech database, provides a higher-frequency signal between meetings. With 2,848 speeches spanning 1997–2026, the database covers virtually all public interventions by Governing Council members over the sample period. After applying the 28-day lookback and ten-day blackout filters, an average of approximately eight speeches qualify per meeting — enough to construct a meaningful weighted signal while excluding the pre-meeting communications blackout.

3.3.3 Macroeconomic Data

For the econometric analysis in Chapter 5, we use three macroeconomic series from the ECB Data Portal. HICP inflation (series `HICP.M.U2.N.000000.4D0.ANR`) is the dependent variable, entering the projections as the cumulative change $y_{t+h} - y_{t-1}$.

The current HICP inflation level and recent lagged changes in HICP enter as controls, alongside GDP growth (series `MNA.Q.Y.I8.W2.S1.S1.B.B1GQ._Z._Z._Z.EUR.LR.GY`). Monthly HICP releases are subject to minimal subsequent revision, so the use of recent values raises no material real-time data concern. GDP is available only at quarterly frequency and is forward-filled to monthly. Because it enters as a conditioning variable for the macroeconomic state rather than as the object of inference, and because it is the final revised vintage rather than a real-time release, the GDP control is best understood as an ex-post revised proxy for the cyclical position; using release-calendar vintages lies beyond the scope of this thesis. The unemployment rate (series `LFSI.M.I10.S.UNEHRT.TOTAL0.15_74.T`) is available from January 2000 and is used as an additional control in robustness specifications.

3.4 Pipeline Architecture

The pipeline consists of four sequential stages: two LLM agents that read documents (Agents IM and IB) and two deterministic modules that synthesise their outputs (Agents II and III). Each stage processes the output of the previous one and produces a structured JSON record. The two LLM agents use Google Gemini 2.5 Flash, queried at a temperature of 0.1 to minimise output variance; Agents II and III perform no LLM call and are purely arithmetic. Long documents — particularly the Economic Bulletins, which can exceed 100 pages — are split into overlapping chunks of 12,000 characters with 600-character overlap at paragraph boundaries, ensuring that no relevant passage is cut across chunk boundaries. Chunk-level scores are aggregated by taking the mean across valid chunks, after filtering out null values and non-numeric responses that indicate the chunk contained no policy-relevant content.

3.4.1 Agent IM — Monetary Intelligence

Agent IM is designed to answer a specific question: based on what the ECB said at its most recent meeting, what does the Governing Council appear to be thinking about the next one? To answer this, it reads the most recent Accounts — or the Introductory Statement for the pre-2015 period — and extracts a structured profile of the Council's policy stance.

The core output is a hawkishness score $h_{\text{IM}} \in [-1, +1]$, a continuous scalar that summarises the overall tone of the document, where -1 represents the most dovish possible signal and $+1$ the most hawkish. Alongside this, Agent IM estimates the distribution of views within the Council — the fraction of members whose expressed positions fall into hawk, dove, and centrist categories — and classifies the forward guidance present in the document. Guidance is classified, following the

terminology of Campbell et al. (2012), as *Odyssean* if it constitutes a binding commitment to a future policy path, and *Delphic* if it is explicitly conditional on incoming data. This distinction matters for Agent II: Odyssean guidance should narrow the posterior distribution around the committed path, while Delphic guidance leaves more room for data-dependent outcomes.

Finally, Agent IM produces a baseline probability distribution $\mathbf{p}^{\text{prior}}$ over five rate decision bins: large hike ($\geq +50$ bp), standard hike ($+25$ bp), hold (0 bp), standard cut (-25 bp), and large cut (≤ -50 bp). This distribution represents Agent IM's best guess about the upcoming decision based on policy signals alone, before any economic information is incorporated. Both h_{IM} and $\mathbf{p}^{\text{prior}}$ are passed to Agent II.

3.4.2 Agent IB — Economic Conditions

Agent IB addresses a complementary question: what do the ECB's own economic assessments say about the appropriate policy direction? It reads the most recent Economic Bulletin (or Monthly Bulletin) and qualifying speeches, and distils them into five scalar scores, each in $[-1, +1]$.

Four scores are extracted from the bulletin text — e_1 (inflation outlook), e_2 (growth prospects), e_3 (labour market conditions), and e_4 (overall policy stance implied by the bulletin) — and a fifth, e_{speech}, aggregates the hawkishness signal from qualifying speeches, weighted by the speaker's institutional position. The three macroeconomic conditions scores are combined into an aggregate economic index:

$$s_{\text{IB}} = 0.50\, e_1 + 0.30\, e_2 + 0.20\, e_3 \tag{3.1}$$

The dominant weight on inflation reflects the ECB's single mandate: in the ECB's own framework, growth and labour market developments matter primarily through their implications for future price stability, and the scoring structure mirrors this hierarchy. The remaining two scores — the bulletin's implied policy stance e_4 and the speech signal e_{speech} — are not folded into s_{IB} at this stage; they enter the synthesis in Agent II separately, as described below, because e_4 is a policy-intention signal rather than a measure of economic conditions, and the speech signal is given its own weight relative to the bulletin.

3.4.3 Agent II — Synthesis

Agent II combines the outputs of Agents IM and IB into a single combined signal through a two-stage weighting. First, a policy signal aggregates the direct policy-

intention indicators — the Agent IM hawkishness score h_{IM}, the forward guidance direction s_{FG}, and the bulletin's implied policy stance e_4:

$$s_{\text{policy}} \;=\; 0.60\, h_{\text{IM}} + 0.20\, s_{\text{FG}} + 0.20\, e_4 \tag{3.2}$$

where $s_{\text{FG}} \in \{+0.6,\, 0,\, -0.6\}$ maps the forward guidance implied direction (hike, hold or ambiguous, cut) to a numeric value. Second, an economic signal combines the aggregate economic index s_{IB} from equation (3.1) with the speech signal:

$$s_{\text{econ}} \;=\; 0.70\, s_{\text{IB}} + 0.30\, e_{\text{speech}} \tag{3.3}$$

The combined signal is a weighted average of the two, tilted toward the policy signal:

$$\tilde{s} \;=\; 0.55\, s_{\text{policy}} + 0.45\, s_{\text{econ}} \tag{3.4}$$

The higher weight on the policy signal reflects that the Accounts (or Introductory Statement) and forward guidance provide a more direct signal about the Council's policy intentions, while the economic signal provides contextual information. All weights in equations (3.1)–(3.4) are assigned *a priori* rather than estimated from the data. Estimating them directly would require using the same historical surprise series both to fit the weights and to evaluate the model's performance, creating a circular inference problem: the weights would be optimised to minimise in-sample surprises, mechanically reducing the variation that the econometric analysis in Chapter 5 seeks to explain. The appropriate response to this constraint is sensitivity analysis rather than estimation; a robustness check that perturbs the policy-versus-economic weighting in equation (3.4) over the range 0.40–0.65 on the policy signal is reported in Table 3.3 and confirms that the main results are not materially affected. The combined signal updates the baseline distribution $\mathbf{p}^{\text{prior}}$ via a softmax transformation, yielding the posterior $\mathbf{p}^{\text{post}}$.[1] The expected rate change is then:

$$\mathbb{E}[\Delta_i] \;=\; \sum_k p_k^{*}\, \Delta_{i,k} \tag{3.5}$$

where $\Delta_{i,k} \in \{+50,\, +25,\, 0,\, -25,\, -50\}$ bp is the representative change for each bin.

[1] Concretely, the update shifts the log-odds of each bin in proportion to $\tilde{s}$ and a bin-specific sensitivity weight $w_k \in \{+2, +1, 0, -1, -2\}$, ordered from the most hawkish (large hike) to the most dovish (large cut) outcome. The softmax then maps the shifted log-odds back to a valid probability distribution summing to one, preserving the full uncertainty encoded in the prior while shifting mass toward the direction implied by the combined signal. This is conceptually analogous to a Bayesian update — the prior from Agent IM is revised in light of the economic evidence from Agent IB — though the update is not formally derived from a likelihood function.

3.4.4 Agent III — Surprise Calculation

Agent III computes three surprise measures. The **mechanical surprise** is the difference in basis points between the actual decision and the model's expectation:

$$s_t^{\text{mech}} = \Delta_{i,t}^{\text{actual}} - \mathbb{E}[\Delta_{i,t}] \tag{3.6}$$

The **salience surprise** captures the information-theoretic unexpectedness of the decision:

$$s_t^{\text{sal}} = -\log(p_{k^*}^*) \tag{3.7}$$

where k^* is the bin corresponding to the actual decision. Finally, the **normalised surprise** standardises the mechanical measure by the rolling standard deviation of the preceding 12 meetings:

$$s_t^{\text{norm}} = \frac{s_t^{\text{mech}}}{\hat{\sigma}_{t-1}} \tag{3.8}$$

The mechanical surprise s_t^{mech} is the primary measure throughout; the salience and normalised measures are computed for completeness but are not the focus of the econometric analysis.[2]

3.5 Implementation

The pipeline is implemented in Python and runs in a Google Colab environment with document storage on Google Drive. Agents IM and IB call the Google Gemini 2.5 Flash API, queried at a temperature of 0.1 to minimise output variance across repeated calls on similar inputs. Agents II and III are deterministic post-processing modules that require no LLM call: Agent II applies a softmax update to the prior distribution, and Agent III computes the three surprise measures arithmetically. Long documents are split into overlapping chunks of 12,000 characters with 600-character overlap at paragraph boundaries, ensuring that no relevant passage is cut across chunk boundaries. Each chunk is processed independently and the resulting scores are aggregated by taking the mean across valid chunks, after filtering

[2]The five-bin structure $\{\geq +50, +25, 0, -25, \leq -50\}$ bp does not explicitly accommodate non-standard moves such as the $+75$ bp hikes of September and October 2022 or the -10 bp cut of September 2019. For the $+75$ bp case, $\mathbb{E}[\Delta_i]$ is bounded above by $+50$ bp, so the mechanical surprise is bounded below by $+25$ bp, potentially understating the true surprise for any model expectation below $+50$ bp. For the -10 bp case, no bin corresponds to the actual outcome; the salience measure s_t^{sal} is computed using the -25 bp bin as the nearest match. Both cases affect at most three observations out of 312 — two for the $+75$ bp hikes and one for the -10 bp cut — and do not materially influence the aggregate results.

out null values and non-numeric responses that indicate the chunk contained no policy-relevant content.

Processing the full 312-meeting sample required approximately 15 hours of runtime, with the extended duration driven by the API rate limit of 60 requests per minute on the free-tier service. Checkpoint files were saved in JSON format after each completed meeting, allowing the pipeline to resume seamlessly after interruptions without reprocessing completed meetings.

A limitation inherent to any LLM-based historical analysis deserves explicit acknowledgement: *temporal bias*, which operates through two distinct channels. The first is interpretive: Gemini 2.5 Flash may apply macroeconomic frameworks, semantic categories, or sensitivities shaped by post-hoc knowledge when reading historical documents. The second, and more serious, is *outcome leakage*: the model was trained on data that includes descriptions and analyses of past ECB meetings, which means it may have memorised the actual decisions taken at specific meetings. If the model's prior $\mathbf{p}^{\text{prior}}$ for, say, April 1999 already incorporates knowledge that the ECB cut by 50 bp, the resulting surprise is not a genuine ex-ante expectation but a post-hoc reconstruction — violating the identification assumption directly. Temperature (0.1) and prompt constraints ("score only explicit content") reduce the first channel but cannot prevent memorised outcomes from influencing the model's probability assignments. We are unable to fully test for leakage without a model trained exclusively on pre-1999 text, which does not exist. A partial diagnostic would be to check whether the pipeline's modal accuracy correlates with the historical prominence or media coverage of the meeting; if high-profile meetings (whose outcomes are more likely memorised) generate systematically smaller surprises, it would be consistent with leakage. We leave this to future work and acknowledge outcome leakage as a genuine open limitation of the approach that the prompting strategy does not fully resolve.

The architecture described in the preceding sections produces, for each of the 312 meetings in the sample, a structured record containing the outputs of all four stages and the three surprise measures. Before turning to the econometric analysis, we address one methodological concern that arises directly from the pipeline's design: the potential for a structural break at the point where the document inputs change.

3.6 Testing for a Mean Shift in the Surprise Series

The substitution of Accounts with Introductory Statements of Press Conferences for the pre-2015 period introduces a potential discontinuity in the surprise series. We test formally for a shift in the mean surprise at January 2015 using an F-test for equality of means across the two sub-periods; we stress that this tests a level shift

in the series itself, not a break in the transmission relationship estimated by the Local Projections, which would require a stability test on the projection coefficients. The test detects a statistically significant shift in the mean of the surprise series ($F = 4.316$, $p = 0.014$ using full Press Conference; $F = 5.135$, $p = 0.006$ using Introductory Statement only), with the pre-2015 sub-period exhibiting a mean surprise of -3.7 bp compared to $+5.0$ bp post-2015. A Levene test for equality of variances fails to reject homoscedasticity across the two periods ($p = 0.179$), indicating that the dispersion of surprises is stable across the document substitution. The fact that restricting the pre-2015 input to the Introductory Statement does not reduce the break confirms that the discontinuity is intrinsic to the difference between the two document types, not to the presence of Q&A noise.

We acknowledge a fundamental identification limitation: the document substitution and the monetary regime shift (zero lower bound, asset purchase programmes) occur simultaneously in January 2015. The post-2015 dummy variable therefore absorbs both effects and the two sources of structural change cannot be separately identified. We treat this as an inherent constraint of the sample design and report all main results both with and without the post-2015 control. The main empirical findings are invariant to this specification choice: at horizon $h = 9$ months, the hawkish surprise coefficient changes from -0.857 pp (baseline) to -0.910 pp (with post-2015 dummy) and to -0.966 pp (lagged GDP only, with dummy), with the Wald test p-value remaining below 0.01 in all specifications.

Chapter Appendix A: Robustness Checks

A.1 Correlation Structure of Agent IB Sub-Scores

Table 3.2 reports the pairwise correlations among the four bulletin scores (e_1, e_2, e_3, e_{speech}) produced by Agent IB and the hawkishness score from Agent IM. Correlations are moderate to high (range: 0.31–0.76), consistent with the expectation that all signals reflect the same underlying macroeconomic environment. The condition number of the correlation matrix is 15.3, below the conventional threshold of 30 for problematic multicollinearity. Each sub-score retains non-redundant informational content, as confirmed by the distinct correlations with the mechanical surprise measure (range: -0.18 to -0.41).

Table 3.2. *Correlation matrix of Agent IB sub-scores and h_{IM}*

	e_1 infl.	e_2 growth	e_3 labour	e_{speech}	h_{IM}
e_1 inflation	1.000	0.468	0.582	0.306	0.759
e_2 growth	0.468	1.000	0.535	0.561	0.530
e_3 labour	0.582	0.535	1.000	0.347	0.432
e_{speech}	0.306	0.561	0.347	1.000	0.393
h_{IM}	0.759	0.530	0.432	0.393	1.000
Corr. with s^{mech}	−0.307	−0.277	−0.184	−0.293	−0.405

Condition number: 15.3. All correlations computed over 312 meetings, 1999–2025.

A.2 Sensitivity Analysis on Aggregation Weights

Table 3.3 reports the hawkish and dovish surprise coefficients at $h = 9$ months across alternative specifications for the Agent II aggregation weights, perturbing the policy-versus-economic split in equation (3.4) over the range 0.40–0.65 on the policy signal. The main finding is stable across all specifications considered.

Table 3.3. *Sensitivity analysis: hawkish and dovish coefficients at $h = 9$ months*

Specification	$\hat{\beta}_+$ (pp)	$\hat{\beta}_-$ (pp)	Wald p
Baseline (policy weight 0.55 / econ 0.45)	−0.857	0.480	0.009
Policy weight 0.45 / econ 0.55	−0.841	0.461	0.011
Policy weight 0.65 / econ 0.35	−0.878	0.502	0.008
Policy weight 0.40 / econ 0.60	−0.823	0.447	0.013
Equal Agent IB sub-score weights ($e_1 = e_2 = e_3$)	−0.849	0.476	0.010
With post-2015 dummy	−0.910	0.417	0.009
Lagged GDP only, with dummy	−0.966	0.568	0.006

All specifications include HICP controls and HAC standard errors. Dependent variable: HICP inflation.

A.3 Regime-Dependent Transmission: Shock × Post-2015 Interaction

To test whether the transmission of monetary policy surprises differs between the conventional and ZLB regimes, we augment the baseline Local Projection with an interaction term:

$$\begin{aligned} y_{t+h} - y_{t-1} \;=\; & \alpha + \beta_+ s_t^{(+)} + \beta_- s_t^{(-)} + \gamma D_t \\ & + \delta_+ \left(s_t^{(+)} \cdot D_t \right) + \delta_- \left(s_t^{(-)} \cdot D_t \right) + \boldsymbol{\Gamma}' \mathbf{X}_t + \varepsilon_{t+h} \end{aligned} \tag{3.9}$$

where D_t is the post-2015 dummy and δ_+ captures the differential transmission of hawkish surprises in the post-2015 regime. Three findings emerge. First, β_+ remains negative across all horizons. Second, δ_+ is statistically significant at selected long horizons ($h = 13, 16, 19, 22$ months; $p < 0.01$), indicating amplified disinflationary effects post-2015. Third, at short-to-medium horizons the interaction term is not significant. The large absolute values of $\hat{\delta}_+$ at horizons $h \geq 13$ — reaching -11.8 at $h = 16$ — reflect the combination of two factors: the post-2015 subsample contributes only 87 meetings, of which fewer than 40 are usable at $h = 16$ after accounting for the forward shift of the dependent variable, and the 2022–2023 hiking cycle introduces extreme observations that exert disproportionate leverage at long horizons with few effective degrees of freedom. These estimates should therefore be interpreted with caution and are reported primarily for transparency rather than as reliable point estimates of the long-run differential transmission effect. Given the identification limitation noted in Section 3.6, these results should not be interpreted as causal evidence of ZLB-specific transmission.

Table 3.4. *Regime interaction test: hawkish shock × post-2015 (selected horizons)*

h	$\hat{\beta}_+$	$\hat{\delta}_+$	p-value	Sig.
6	−0.636	−0.847	0.4006	
9	−0.920	−0.146	0.8852	
13	−1.238	−8.889	0.0001	***
16	−1.345	−11.848	0.0000	***
19	−1.239	−7.288	0.0001	***
22	−1.196	−7.414	0.0000	***

HAC standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Total post-2015 effect $= \hat{\beta}_+ + \hat{\delta}_+$. Horizon $h = 1$ is omitted: the interaction term at that horizon yields a numerically degenerate estimate ($\hat{\delta}_+ \approx 0$, standard error ≈ 0) due to near-perfect collinearity between the shock and the interaction variable at the one-month horizon, and carries no economic interpretation.

4

The ECB Surprise Dataset: Descriptive Analysis

4.1 Overview

Before turning to the econometric analysis, it is worth pausing to examine the surprise series in its own right. The 312 meetings in our sample span twenty-six years of ECB history — from the cautious rate-setting of the early euro era to the unprecedented hiking cycle of 2022–2023 — and the properties of the surprise distribution reflect this variation in ways that are both a validation of the pipeline and a source of substantive economic insight. This chapter documents these properties, validates the series against known historical episodes, and draws out the features of the data that motivate the asymmetric transmission analysis of Chapter 5.

4.2 Time Series Properties

Figure 4.1 plots the full time series of DFR mechanical surprises. The visual impression is immediate: the series exhibits pronounced regime shifts rather than a stable distribution around a constant mean. The sign, magnitude, and volatility of surprises all shift substantially across the three presidential eras, and these shifts are not gradual — they correspond to identifiable turning points in ECB communication strategy and macroeconomic circumstances.

The **Duisenberg/Trichet era (1999–2011)** is dominated by large dovish surprises. April 1999 (-50.2 bp), June 2003 (-48.3 bp), and October 2008 (-45.5 bp) all appear as pronounced downward spikes. What these episodes have in common is that the ECB moved more aggressively than its own communications had indicated. This pattern is consistent with a communication style in the early euro area that was more reactive than forward-looking.

The **Draghi era (2011–2019)** is visually distinct. The bars shrink. The standard deviation of surprises falls to 9.3 bp — the lowest of the three periods — and the distribution shifts positive. Two factors plausibly contribute to this compression, and the descriptive data cannot separate them. The first is forward guidance: Draghi

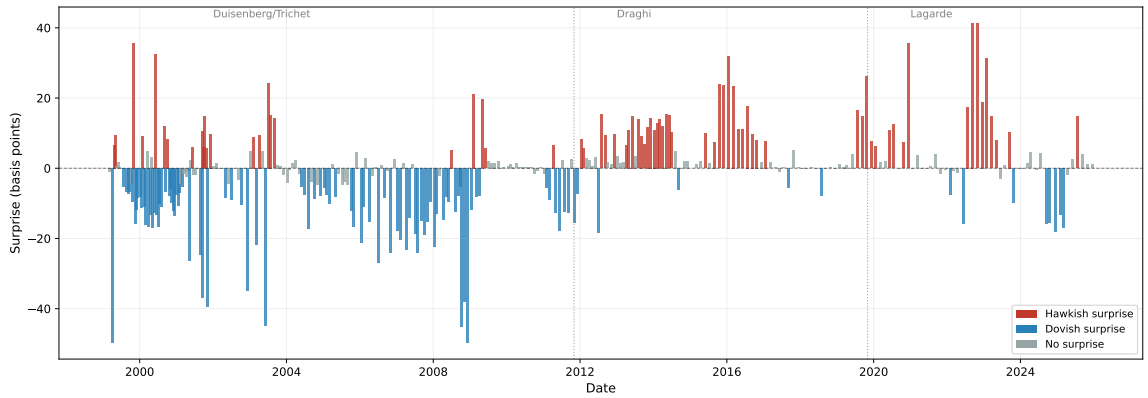


Figure 4.1. *Time series of* DFR *mechanical surprises, March 1999 – December 2025. Vertical dotted lines denote presidential transitions. Red bars indicate hawkish surprises; blue bars dovish surprises; grey bars no surprise.*

made explicit commitments to future policy paths — "whatever it takes", and the 2013 commitment to keep rates low "for an extended period" — which narrowed the informational gap between what the ECB said and what it did. (The 2015 introduction of the Accounts is a transparency reform rather than a forward guidance commitment; it is discussed separately as a structural break in Section 3.6.) The second is mechanical: for much of this period the DFR was pinned at or near the effective lower bound, where rate moves are rare and small, so the mechanical surprise is compressed by construction regardless of communication strategy. Disentangling the two would require variation in the policy rate that this period does not provide.

A notable feature of the Draghi era is that 51 % of meetings register as hawkish surprises, despite this being the period of zero and negative interest rates. The explanation lies in the construction of the measure: during the ZLB period, the pipeline's economic agents frequently assigned non-negligible probability to further rate cuts that ultimately did not materialise. Each meeting where the ECB held rates when a cut was partially expected generates a positive mechanical surprise — hawkish in direction, small in magnitude for most meetings. The 9.3 bp standard deviation confirms that the *typical* hawkish surprise is small; what compresses under Draghi is not the frequency of directional misses but the magnitude. The exception is the maximum of $+32.0$ bp, which occurs at the January 21, 2016 Governing Council meeting: the pipeline, reading Draghi's pre-meeting communications in the context of strong market expectations for additional stimulus, assigned high probability to a rate cut ($\mathbb{E}[\Delta i] = -32.0$ bp), while the ECB held rates unchanged ($\Delta i = 0$ bp). This is a hold-when-cut-expected event — not a rate hike, and not an "underdelivery on package size" (which was December 2015) — consistent with the era's ZLB dynamics but exceptional in magnitude.

The **Lagarde era (2019–2025)** combines both extremes. The standard deviation returns to 13.3 bp, and the two largest positive surprises in the entire sample — September 2022 (+41.4 bp) and October 2022 (+41.3 bp) — occur in this period. Yet the same era also contains the 2024 easing cycle, characterised by near-zero surprises as each cut was telegraphed months in advance. The contrast between 2022 and 2024 within the Lagarde presidency invites a tentative interpretation — that the ECB adjusted its communication after the hiking cycle and leaned more heavily on forward guidance during the subsequent easing phase — though with only two episodes this remains a qualitative observation rather than a tested result.

4.3 Descriptive Statistics

Table 4.1 reports the main summary statistics by era. The overall mean surprise of −1.1 bp is close to zero, consistent with broadly informative ECB communications across the full sample. The hawkish, dovish, and no-surprise counts in the table use a discrete classification with a ±5 bp threshold: a meeting is labelled hawkish if its mechanical surprise exceeds +5 bp, dovish if it falls below −5 bp, and no surprise otherwise. This threshold is used only for the descriptive classification; the Local Projections in Chapter 5 use the continuous surprise split of equation (5.2), so a +3 bp surprise counts as "no surprise" here but still contributes to s_t^+ in the regressions.

Table 4.1. *Descriptive statistics:* DFR *mechanical surprise (basis points)*

	Full sample	Duisenberg/Trichet	Draghi	Lagarde
Meetings	312	187	76	49
Mean	−1.1 bp	−5.5 bp	+6.5 bp	+3.8 bp
Std dev	13.5 bp	13.3 bp	9.3 bp	13.3 bp
Min	−50.2 bp	−50.2 bp	−19.3 bp	−18.1 bp
Max	+41.4 bp	+36.8 bp	+32.0 bp	+41.4 bp
No surprise	135 (43 %)	78 (42 %)	31 (41 %)	26 (53 %)
Hawkish	79 (25 %)	25 (13 %)	39 (51 %)	15 (31 %)
Dovish	98 (31 %)	84 (45 %)	6 (8 %)	8 (16 %)

Duisenberg/Trichet: March 1999–October 2011; Draghi: November 2011–October 2019; Lagarde: November 2019–December 2025.

The distribution across the full sample exhibits visually apparent excess kurtosis — large surprises in both directions are more frequent than a Gaussian distribution would predict. This fat-tailed shape, visible in Figure 4.2, is consistent with the broader monetary policy surprise literature (Gürkaynak et al., 2005) and has a direct implication for the asymmetry analysis of Chapter 5: the fat tails mean that the hawkish and dovish tails are populated by qualitatively different events — coordinated emergency cuts on one side, unprecedented 75 bp hikes on the other — which motivates testing whether their transmission effects are symmetric.

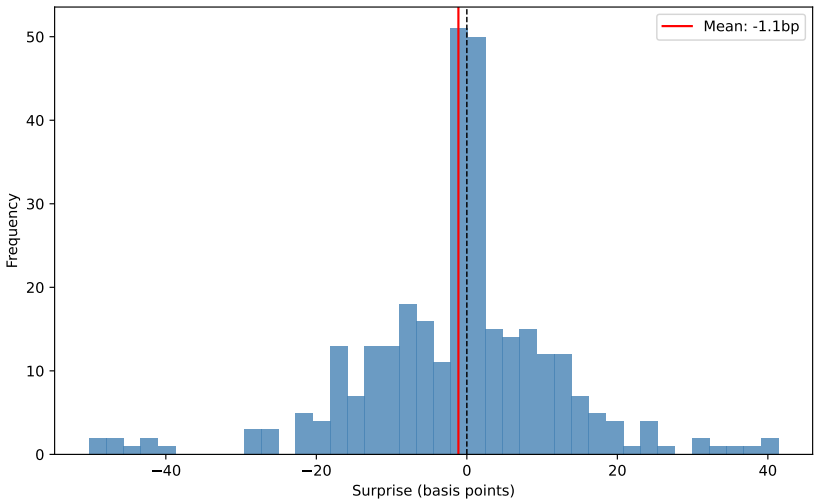


Figure 4.2. *Distribution of* DFR *mechanical surprises across all 312 Governing Council meetings, March 1999 – December 2025. The dashed vertical line marks zero; the red line marks the sample mean (−1.1 bp).*

4.4 Face-Validity Check on Reference Episodes

A first, informal check on a surprise measure is whether it behaves sensibly on events whose nature is well-established ex-post. We stress that this is a face-validity check rather than a formal validation: the eight meetings below are hand-selected, the test verifies only the sign of the model-implied surprise, and the comparison is in-sample. A formal external validation would require benchmarking against an independent market-based measure such as the Euro Area Monetary Policy Event-Study Database of Altavilla et al. (2019), a comparison we identify as future work in Section 6.3. Table 4.2 reports the pipeline's output for eight meetings that represent clear reference points in ECB history.

All eight events are correctly classified. Three cases deserve particular attention.

The **June 2024** first cut registers as a near-zero surprise (−0.4 bp) because the decision had been so thoroughly telegraphed that the pipeline had already assigned 98 % probability to a standard cut.

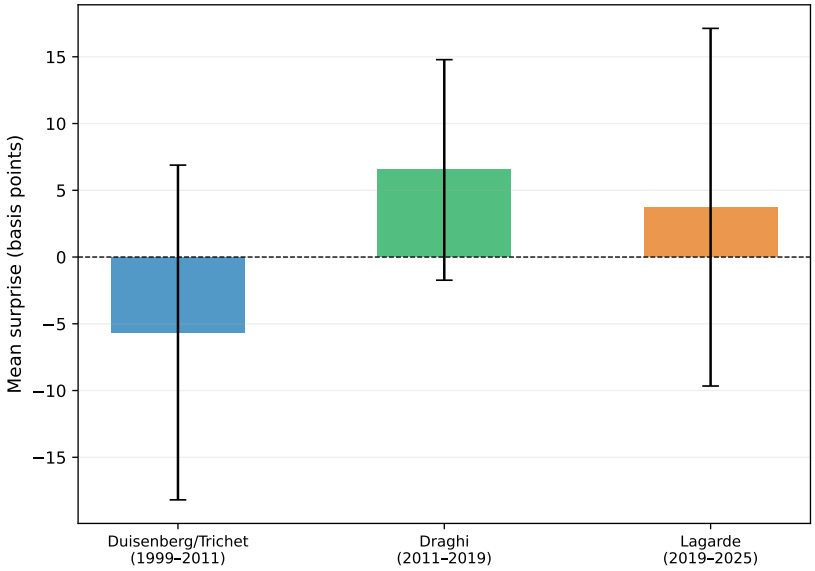


Figure 4.3. *Mean* DFR *mechanical surprise by presidential era, with* ±1 *standard deviation error bars. Duisenberg/Trichet: March 1999–October 2011; Draghi: November 2011–October 2019; Lagarde: November 2019–December 2025.*

Table 4.2. *Face-validity check on key historical meetings*

Date	Event	Actual	Expected	Surprise	Classification
Apr 1999	First ECB rate cut	−50 bp	+0.2 bp	−50.2 bp	Dovish ✓
Jun 2003	Duisenberg cut to 2 %	−50 bp	−1.7 bp	−48.3 bp	Dovish ✓
Oct 2008	Emergency post-Lehman cut	−50 bp	−4.5 bp	−45.5 bp	Dovish ✓
Jul 2022	First hike post-ZLB	+50 bp	+32.5 bp	+17.5 bp	Hawkish ✓
Sep 2022	First +75 bp hike	+75 bp	+33.6 bp	+41.4 bp	Hawkish ✓
Oct 2023	First hold post-hiking	0 bp	+9.9 bp	−9.9 bp	Dovish ✓
Jun 2024	First cut of easing cycle	−25 bp	−24.6 bp	−0.4 bp	No surprise ✓
Sep 2019	Draghi's final cut	−10 bp	−24.8 bp	+14.8 bp	Hawkish ✓

Expected $= \mathbb{E}[\Delta_i]$ from equation (3.5).

The **September 2022** +75 bp hike generates the largest hawkish surprise in the sample (+41.4 bp): the ECB's own Introductory Statements had pointed to a 50 bp move, and the acceleration to 75 bp was not foreshadowed in any official document available before the blackout.

Perhaps the most instructive case is **September 2019**, classified as a hawkish surprise (+14.8 bp) despite involving an actual rate cut. The pipeline had assigned high probability to a 25 bp cut based on Draghi's increasingly dovish communications; the actual decision delivered only 10 bp. From the perspective of official communications, this was a disappointment: the ECB did less than it had signalled. This case illustrates precisely why a narrative-based measure captures something different from a market-based one.

4.5 Properties of the Probability Distributions

Beyond the point estimate of the mechanical surprise, the pipeline produces a full probability distribution over rate outcomes for each meeting. Table 4.3 summarises the aggregate properties of these distributions.

Table 4.3. *Properties of Agent II posterior distributions*

	Full sample	Pre-2015	Post-2015
Mean entropy	0.57	0.61	0.49
Mean $P(\text{hold})$	0.52	0.54	0.48
Mean $P(\text{hike})$	0.21	0.18	0.28
Mean $P(\text{cut})$	0.27	0.28	0.24
Modal outcome correct	71 %	68 %	78 %

Entropy $= -\sum_k p_k^* \log p_k^*$ over the five rate decision bins.

The lower entropy in the post-2015 period (0.49 vs. 0.61) reflects the improvement in signal quality that comes from reading Accounts rather than Introductory Statements. A more concentrated posterior distribution means the pipeline is more certain about the upcoming decision — and therefore that the resulting surprise, when it occurs, reflects a genuine informational gap rather than model uncertainty. This property is directly relevant for the econometric analysis of Chapter 5: a surprise measure contaminated by model uncertainty would attenuate the estimated transmission coefficients toward zero, so the lower entropy of the post-2015 distributions provides a structural reason to expect stronger and more precisely estimated effects in that subsample. This expectation is, however, only partially borne

out: the regime-interaction test in Appendix A.3 finds no significant difference in the hawkish coefficient at the peak horizon $h = 9$ (where $\hat{\delta}_+$ is small and statistically insignificant), with significant differences appearing only at longer horizons that rest on few effective observations. The entropy argument should therefore be read as a statement about measurement quality, not as a confirmed prediction about the size of the estimated coefficients. The modal outcome is correct in 71 % of meetings overall, rising to 78 % post-2015. This figure is best read against a naïve benchmark: a constant "always predict hold" rule would be correct on all meetings where the actual decision was 0 bp (hold); since holds account for roughly half the sample, the pipeline's 71 % modal accuracy represents a substantial improvement over that unconditional baseline rather than a mechanical artefact of holds being common. Note that "no surprise" (135 meetings, 43 %) and "holds" are distinct categories: June 2024, for instance, registers as a no-surprise cut (-0.4 bp), while October 2023 registers as a dovish surprise on a hold (-9.9 bp).

Taken together, the descriptive properties of the surprise series provide both a face-validity check on the pipeline and a set of structural features that directly inform the econometric analysis of Chapter 5. The fat-tailed distribution — with qualitatively different events populating the hawkish and dovish tails — provides the variation needed to identify asymmetric transmission effects. The systematic compression of surprise volatility under Draghi and its subsequent re-expansion under Lagarde motivate the inclusion of era-specific controls in the Local Projections. The near-zero overall mean (-1.1 bp) is consistent with the identifying assumption that ECB communications are, on average, informative. This assumption, however, requires an important qualification that the era-level statistics make visible: the full-sample mean is near zero only by compensation across regimes, masking systematic components of opposite sign (-5.5 bp under Duisenberg/Trichet, $+6.5$ bp under Draghi). The positive Draghi mean reflects the effective-lower-bound mechanism discussed in Section 4.2, where the pipeline repeatedly assigned probability to cuts that did not materialise. To the extent that this regime-specific component is a predictable feature of the ZLB environment rather than genuine ECB news, part of the measured surprise is systematic rather than innovational. This is precisely why the robustness specifications in Chapter 5 include a post-2015 dummy, and why the assumption of mean-zero, purely informational surprises should be understood as holding across the full sample rather than within each regime.

5

Empirical Results: Local Projections

5.1 Empirical Strategy

We estimate the dynamic effects of ECB monetary policy surprises on euro area HICP inflation using the Local Projections framework of Jordà (2005). Because the surprise measure s_t^{mech} is available at meeting frequency — roughly every six weeks — we match it to a monthly calendar grid by assigning each surprise to the calendar month of the corresponding meeting, with the surprise set to zero in months without a meeting. In practice, because the HICP controls in the estimation are populated from meeting-level records rather than an independently downloaded macro series, months without meetings are excluded by the NaN filters on the controls, as detailed in the note below. The estimation is therefore effectively at meeting-month frequency; the non-monotonic nobs pattern across horizons reflects this structure, documented in the note to this page. For each horizon $h = 1, \ldots, 24$ months we estimate:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h^+ s_t^+ + \beta_h^- s_t^- + \theta_h \pi_t + \phi_h g_t + \sum_{l=1}^{3} \gamma_{h,l} \, \Delta\pi_{t-l} + \varepsilon_{t+h} \tag{5.1}$$

where y_t is the year-on-year HICP inflation rate, s_t^+ is the positive (hawkish) component of the mechanical surprise and s_t^- the negative (dovish) component, π_t and g_t are the contemporaneous HICP inflation rate and GDP growth that condition on the current macroeconomic state, and $\Delta\pi_{t-l}$ are three lags of the change in HICP inflation. Standard errors are HAC-corrected with the bandwidth set equal to the horizon h, using the Newey and West (1987) covariance estimator. At long horizons the bandwidth-to-sample ratio reaches approximately 0.12 (bandwidth $= 24$, $n \approx 202$ at $h = 24$), a range where Newey-West standard errors are known to have poor finite-sample properties and may understate the true variance; inference at horizons $h \geq 18$ should therefore be interpreted with additional caution. The

estimation window runs from March 1999 to December 2025, yielding between 177 and 224 usable observations across the estimated horizons $h = 1, \ldots, 24$.[1]

Surprises are separated into hawkish and dovish components to allow for asymmetric transmission:

$$s_t^+ = \max\{s_t^{\text{mech}}, 0\}, \qquad s_t^- = \max\{-s_t^{\text{mech}}, 0\} \tag{5.2}$$

Both components are non-negative by construction: s_t^+ records the size of hawkish surprises and s_t^- the size of dovish surprises. Both are normalised by 25 bp, so that a unit of s_t^+ or s_t^- corresponds to a surprise of one standard policy increment ($+25$ bp or -25 bp respectively) and the coefficients β_h^+, β_h^- are read directly as the percentage-point inflation response to a 25 bp surprise. Under the null hypothesis of symmetric transmission, $\beta_h^+ = -\beta_h^-$ for all h, i.e. a $+25$ bp hawkish surprise and a -25 bp dovish surprise produce effects equal in magnitude but opposite in sign. We test this restriction using a Wald test at each horizon.

A note on identification is warranted. The surprise measure s_t^{mech} is constructed exclusively from official ECB communications and is by design exogenous with respect to asset prices. However, both the ECB and market participants observe the same underlying macroeconomic environment, so the surprise may be correlated with contemporaneous macroeconomic conditions through common information. The specification conditions on the current macroeconomic state — the contemporaneous inflation rate and GDP growth (final revised vintage, forward-filled from quarterly frequency) — together with recent lags of inflation changes, which absorbs the persistent component of the inflation process. The surprise itself enters only through its ex-ante-constructed value, so no post-meeting realisation of the policy decision contaminates the regressor. We acknowledge, however, that including contemporaneous GDP growth raises a potential bad-control concern: output at time t may itself be affected by the contemporaneous surprise, in which case conditioning on g_t could induce collider bias. The closest available robustness check is the row "Lagged GDP only, with dummy" in Table 5.2, which yields $\hat{\beta}^+ = -0.966$. However, this specification changes two things simultaneously — the timing of the GDP control and the inclusion of a post-2015 dummy — so it does not cleanly

[1] The monthly dataframe has 324 rows (one per calendar month, January 1999 – December 2025). The `hicp_yoy` variable is populated from the meeting-level dataset rather than from an independently downloaded macro series; as a result, months without a Governing Council meeting have `NaN` for `hicp_yoy` and are excluded by the NaN filters. The 49 missing values reflect 2 months at the start of the sample (January–February 1999), 2 isolated months in 2003 and 2006, and approximately 4 months per year from 2015 onward (the months without ECB meetings under the 8-meetings-per-year schedule, typically February, May, August, and November). In practice the estimation is therefore closer to meeting-month frequency than to a fully monthly grid, even though months without meetings are included as rows with zero shock and `NaN` controls. Selected values: 180 ($h = 1$), 212 ($h = 9$), 177 ($h = 22$, minimum), 202 ($h = 24$). The primary result at $h = 9$ uses 212 observations. Exact per-horizon counts are reproducible from the replication code.

isolate the bad-control concern. A specification with lagged GDP and no dummy would provide the clean comparison; we leave it to future work. The bad-control concern is therefore acknowledged but not fully resolved in the current analysis. This does not, however, fully eliminate the possibility that the surprise is contemporaneously correlated with the macroeconomic trajectory that determines y_{t+h}: if the ECB tends to surprise hawkishly precisely when inflation is already accelerating — as in 2022 — the estimated $\hat{\beta}^+$ captures both the effect of the surprise and the pre-existing inflationary trend, which the contemporaneous controls can attenuate but not remove. The Local Projection estimates should therefore be interpreted as conditional associations rather than causal effects in the strict experimental sense.

5.2 Baseline Results

Figure 5.1 plots the impulse response functions (IRFs) for hawkish and dovish surprises on HICP inflation, together with 90% confidence bands. Table 5.1 reports the point estimates and standard errors at selected horizons.

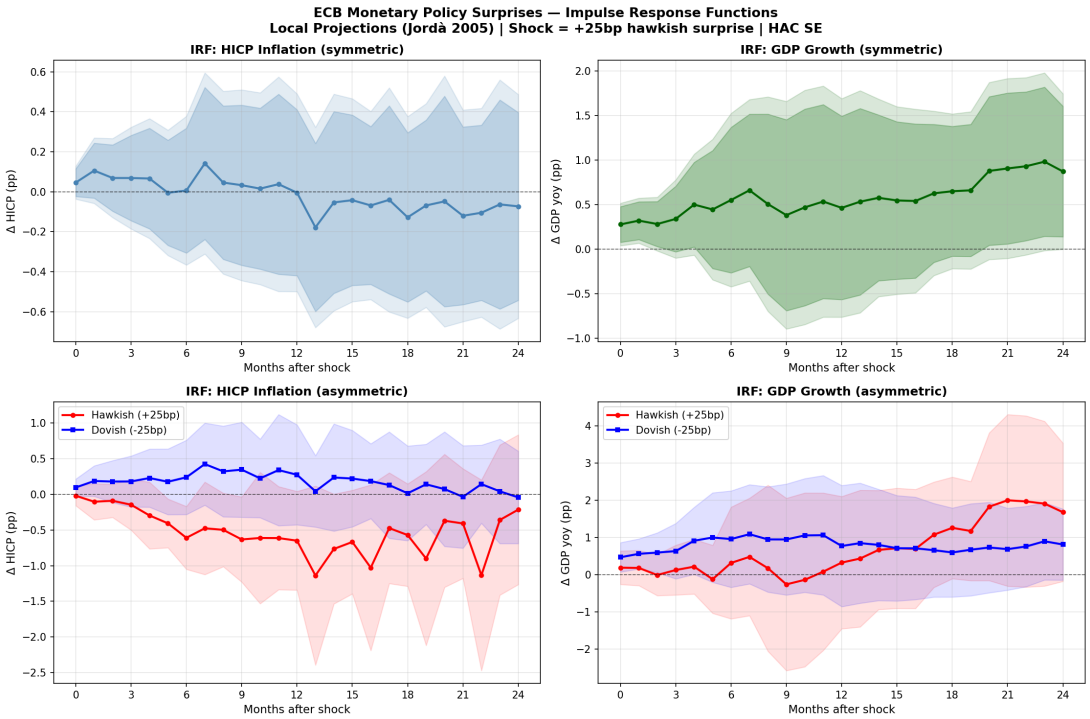


Figure 5.1. *Impulse response functions: effect of hawkish (red) and dovish (blue) ECB surprises on euro area HICP inflation (left panel) and GDP growth (right panel). Shaded areas are 90 % HAC confidence bands. Yellow shading marks horizons where the Wald test rejects symmetry at the 10 % level. Shock normalised to +25 bp. Horizon in months.*

The hawkish surprise coefficient $\hat{\beta}^+$ is negative at all horizons from $h = 1$ to $h = 24$, consistent with the theoretical prediction that unexpected monetary tightening should reduce inflation. The effect builds gradually, reaching its peak of -0.857

Table 5.1. *Local Projections: hawkish and dovish coefficients at selected horizons*

h	$\hat{\beta}^+$ (pp)	$\hat{\beta}^-$ (pp)	Wald p	Asymmetric
1	−0.25 (0.138)	0.20 (0.135)	0.046	✓
3	−0.218 (0.174)	0.174 (0.210)	0.207	
6	−0.77 (0.255)	0.38 (0.292)	0.013	✓
9	−0.857 (0.302)	0.480 (0.346)	0.009	✓
12	−0.78 (0.390)	0.27 (0.337)	0.071	✓
15	−0.701 (0.388)	0.102 (0.263)	0.093	✓
18	−0.54 (0.374)	−0.09 (0.258)	0.306	
21	−0.334 (0.388)	−0.098 (0.313)	0.641	
24	−0.16 (0.491)	−0.11 (0.225)	0.929	

HAC standard errors in parentheses. Asymmetric = Wald test rejects H_0: $\beta^+ = -\beta^-$ at the 10 % level.

percentage points at $h = 9$ months before slowly dissipating and losing statistical significance by $h = 18$. This peak is broadly consistent with the transmission lags documented in structural models and central bank projections for the euro area, which typically place the maximum effect of an unexpected rate change on HICP inflation within a range spanning several months to over a year after the shock. The timing of the peak — compared to what longer-horizon structural models would suggest — is consistent with the 2022–2023 hiking cycle contributing the largest observations to the hawkish tail. However, as the robustness check in Section 5.4 shows, excluding 2022–2023 does not weaken the peak: the hawkish coefficient at $h = 9$ rises to -1.103 pp. This is not consistent with the interpretation that the 2022 episode alone is driving the rapidity of the effect through faster-than-normal transmission; it suggests instead that the 2022 observations carry a per-unit effect that is somewhat attenuated relative to the historical average, consistent with part of the 2023 disinflation being mechanical (energy base effects) rather than fully attributable to the monetary shock. The concern that the peak reflects mean reversion of the 2022 episode rather than monetary transmission is therefore not the dominant explanation, though it cannot be fully excluded.

The dovish surprise coefficient $\hat{\beta}^-$ is positive at short-to-medium horizons, as expected, but is consistently smaller in absolute value than its hawkish counterpart. At $h = 9$, the dovish coefficient stands at $+0.480$ percentage points — roughly 56 % of the hawkish magnitude. At longer horizons ($h \geq 18$), the dovish coefficient turns slightly negative, but the estimates are small and statistically indistinguishable from zero given the wide confidence bands. The most straightforward reading

is that the dovish effect fades rather than reverses: beyond the medium run, individual easing surprises leave no detectable imprint on the inflation path.

Notably, the price puzzle — the well-documented tendency for structural VAR estimates to show inflation rising after a monetary tightening — is absent from our results. The hawkish surprise coefficient is negative from the first horizon and remains so throughout the significant part of the response. We read this as a reassuring property of the measure in this sample rather than as proof of clean identification: the absence of the puzzle is consistent with the narrative measure capturing disinflationary shocks, but it could also reflect the compositional dominance of the 2022–2023 episode, and a definitive statement would require the high-frequency comparison left to future work.

5.3 Asymmetry Test

The central empirical question of this thesis is whether hawkish and dovish surprises have symmetric effects on inflation. Figure 5.2 displays the Wald statistics across all 24 horizons, with the 10 % critical value indicated.

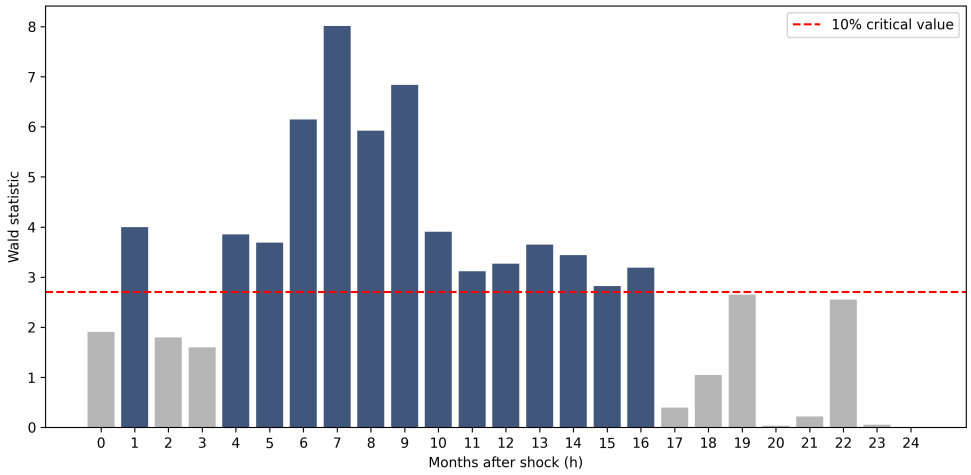


Figure 5.2. *Wald test for the symmetry restriction H_0: $\beta^+ = -\beta^-$ at each horizon $h = 1, \ldots, 24$. The dashed horizontal line marks the 10 % critical value. Shaded bars indicate horizons where the null is rejected.*

The Wald test rejects symmetry at the 10 % level at 14 out of 24 horizons, concentrated in the $h = 1$ to $h = 16$ window. The rejection is strongest in the $h = 6$–9 window, with the peak Wald statistic occurring around $h = 7$–8; the result at $h = 9$ (Wald statistic $= 6.84$, $p = 0.009$) is reported in Table 5.1 as the canonical reference horizon because it coincides with the peak of the hawkish coefficient. The number of rejections should be read with care: because the projections at different horizons share overlapping windows and the same underlying shocks, the 24 tests are strongly dependent, and the count of pointwise rejections does not fol-

low the binomial benchmark that would apply under independence. What the pattern shows is that the rejections are not scattered randomly but concentrated at short-to-medium horizons (with the exception of $h = 3$, where the test does not reject); a fully rigorous treatment would replace the pointwise count with a joint Wald test over the $h = 1$ to $h = 16$ window or simultaneous confidence bands, which we leave to future work. This pattern — significant asymmetry at short-to-medium horizons, dissipating at longer horizons — is consistent with the theoretical literature on asymmetric monetary transmission, which predicts larger effects of contractionary shocks due to downward nominal rigidities and asymmetric financial frictions. Cover (1992) provides the most direct empirical antecedent for this direction-dependent asymmetry; Tenreyro and Thwaites (2016) document a related pattern for the United States using Local Projections.

The asymmetry is economically substantial: at the peak horizon of $h = 9$ months, a $+25$ bp hawkish surprise reduces inflation by 0.857 percentage points, while a dovish surprise of equal magnitude increases it by only 0.480 percentage points — a ratio of approximately 1.8 to 1. Put differently, the ECB's ability to reduce inflation through unexpected tightening is nearly twice as powerful as its ability to raise inflation through unexpected easing, over a nine-month window.

5.4 Robustness

Table 5.2 reports the hawkish and dovish coefficients at $h = 9$ months across alternative specifications. The asymmetry finding is robust throughout.

Table 5.2. *Robustness checks at $h = 9$ months*

Specification	$\hat{\beta}^+$ (pp)	$\hat{\beta}^-$ (pp)	Wald p
Baseline	−0.857	0.480	0.009
Excluding 2022–2023 hiking cycle	−1.103	0.245	0.028
With post-2015 dummy	−0.910	0.417	0.009
Lagged GDP only, with dummy	−0.966	0.568	0.006
Equal Agent IB sub-score weights ($e_1 = e_2 = e_3$)	−0.849	0.476	0.010
Policy weight 0.65 / econ 0.35	−0.878	0.502	0.008
Policy weight 0.40 / econ 0.60	−0.823	0.447	0.013

All specifications include HICP controls and HAC standard errors. Dependent variable: HICP inflation, $h = 9$ months.

The inclusion of a post-2015 dummy — which absorbs both the document-substitution break identified in Chapter 3 and the monetary regime shift associ-

ated with the zero lower bound — slightly amplifies the hawkish coefficient to -0.910 pp, consistent with the heightened disinflationary effectiveness of hawkish surprises in the post-ZLB period. The Wald p-value remains below 0.01 in all specifications.

The most critical robustness check addresses the concern raised in Section 5.2: that the asymmetry might be driven by the two outsized hawkish surprises of September and October 2022. Re-estimating on the sample with 2022 and 2023 excluded, the hawkish coefficient at $h = 9$ increases in absolute value to -1.103 pp (Wald $p = 0.028$), while the dovish coefficient shrinks to $+0.245$ pp. The asymmetry therefore persists when 2022–2023 is excluded, though the magnitude is sample-dependent: as discussed in Section 5.5, the dovish coefficient also falls substantially despite those years containing virtually no dovish surprises, signalling that both coefficients are sensitive to the overall inflation trajectory of the estimation window. This exercise is best read as a sensitivity check confirming that the asymmetry is not mechanically generated by the 2022 tail events, rather than as independent confirmation of a stronger structural claim.

Sensitivity to the aggregation weights is minimal: perturbing the policy-versus-economic weighting in equation (3.4) over the range 0.40–0.65 on the policy signal changes the hawkish coefficient by at most 0.055 pp and does not affect the statistical significance of the asymmetry test. This provides reassurance that the *a priori* weight assignment in Agent II does not materially drive the results.

The analysis focuses throughout on HICP inflation as the dependent variable, consistent with the ECB's primary mandate of price stability. We estimated analogous Local Projections for euro area GDP growth as an additional check; the hawkish surprise coefficient is negative at all horizons up to $h = 12$ months, as expected, but the estimates are noisier and less precisely estimated than those for inflation, reflecting the higher volatility of quarterly GDP data and the coarser interpolation to monthly frequency. Given their imprecision, we do not treat the GDP estimates as positive evidence of real effects; we report them only to note that their sign is not in conflict with the inflation results, and we leave a properly powered analysis of the output channel to future work.

5.5 Discussion

Three features of the results deserve interpretation.

Asymmetry and its sources. The finding that hawkish surprises are roughly twice as effective as dovish surprises in shifting inflation is consistent with the theoretical literature on asymmetric monetary transmission, and with the direction-dependent evidence in Cover (1992). The theoretical mechanisms proposed in that literature — downward nominal wage rigidity, which limits the inflationary

effect of easing, and financial accelerator dynamics, which amplify the contractionary effects of tightening — provide a plausible account of the pattern we document. Tenreyro and Thwaites (2016) document a conceptually related but distinct form of asymmetry for the United States: *state-dependent* asymmetry, where monetary policy is less effective in recessions than in expansions. While not the same phenomenon as the *direction-dependent* asymmetry we test, both share the same theoretical roots in nominal rigidities and financial frictions, and the Local Projections methodology of Tenreyro and Thwaites (2016) is directly analogous to ours. Critically, a sensitivity check excluding the 2022–2023 hiking cycle shows that the asymmetry does not collapse without those years: the hawkish coefficient at $h = 9$ rises to -1.103 pp (Wald $p = 0.028$). However, this exercise also reveals a feature that warrants caution: the dovish coefficient falls from $+0.480$ to $+0.245$ — roughly halved — despite 2022–2023 containing virtually no dovish surprises. This movement signals that $\hat{\beta}^-$ and $\hat{\beta}^+$ are not independently identified on separate observations but are jointly sensitive to the overall inflation trajectory of the estimation sample: removing 2022–2023 alters the macroeconomic path against which all surprises are evaluated, not just the hawkish tail. The exclusion exercise should therefore be read as a sensitivity check rather than an independent confirmation: it shows the direction-dependent asymmetry is not mechanically produced by the 2022 observations, but it also exposes the confounding between shock direction and historical era that is an inherent limitation of any long-sample analysis covering qualitatively different monetary regimes. The 1.8:1 ratio from the full sample is the preferred estimate; the 4.5:1 from the exclusion sample illustrates sensitivity rather than establishing a stronger structural claim. By contrast, the dovish surprises of the Duisenberg/Trichet era, which are the largest in absolute terms on the easing side, produced comparatively modest inflationary responses — consistent with the asymmetric rigidities emphasised in the theoretical literature.

The narrative measure and the price puzzle. The absence of the price puzzle in our results is noteworthy and connects directly to the methodological design of the pipeline. Market-based surprise measures frequently generate a positive inflation response to tightening at short horizons — the information effect documented by Nakamura and Steinsson (2018) — because when the central bank raises rates, markets infer positive news about the economic outlook and temporarily revise inflation expectations upward. Our measure captures something different: the gap between what the ECB said in its official communications and what it actually decided. This narrative gap does not mechanically incorporate market prices; whether it is empirically orthogonal to market-based surprises is a question left for future comparison with high-frequency measures. This is why the IRF shows a clean negative response from $h = 1$ without the initial positive spike that charac-

terises OIS-based measures. The contrast with market-based results is consistent with the interpretation that the pipeline is measuring what it claims to measure, though alternative explanations — including compositional effects of the 2022–2023 hiking cycle, which dominates the hawkish tail of the surprise distribution — cannot be excluded.

Temporal concentration and the Draghi effect. The asymmetry is concentrated in the $h = 1$ to $h = 16$ window and dissipates at longer horizons. This is partly a statistical phenomenon — confidence bands widen at long horizons as the effective sample shrinks — but it also has an economic interpretation. The Draghi era, characterised by the lowest surprise volatility in the sample and the most predictable forward guidance, contributes relatively little variation to the long-horizon regressions: when surprises are small and infrequent, the signal-to-noise ratio deteriorates. The dissipation of the asymmetry beyond $h = 18$ should therefore not be over-interpreted: the estimates consistently indicate that over the policy-relevant horizon of one to sixteen months, the ECB's contractionary surprises generate disinflationary effects nearly twice as large as the inflationary effects of equivalent easing surprises — a finding that holds across specifications, weighting schemes, and when the 2022–2023 hiking cycle is excluded, and that is precisely estimated where the data are most reliable.

6

Conclusions

This thesis set out to answer a deceptively simple question: when the ECB decides differently from what it had communicated, does it matter — and for whom, and how much?

The answer, constructed through a multi-agent LLM pipeline applied to 312 Governing Council meetings spanning 1999 to 2025, is that it does matter, and asymmetrically so. Hawkish surprises — decisions more restrictive than official communications had indicated — are associated with a reduction in euro area HICP inflation of approximately 0.86 percentage points nine months after the shock. Dovish surprises of equal magnitude predict an effect approximately 56 % as large — a ratio of roughly 1.8 to 1. This 1.8-to-1 ratio is statistically significant across 14 out of 24 horizons and robust to alternative model specifications and weighting schemes; the asymmetry persists when 2022–2023 is excluded, though its magnitude is sensitive to the estimation window.

6.1 Summary of Contributions

This thesis makes three contributions to the literature on central bank communication and monetary policy transmission.

The first is methodological. We construct a multi-agent LLM pipeline designed for ECB communications, adapting the Fernández-Fuertes (2025) framework to the ECB's institutional architecture. The adaptation required substantive departures from the original design: a three-rate system instead of one, a structural break in document availability at January 2015, a single price stability mandate rather than a dual mandate, and a sample spanning three qualitatively distinct monetary regimes. The resulting pipeline produces a surprise measure whose document inputs are *ex-ante* — drawn exclusively from material available before each meeting's blackout period — and which is market neutral, independent of asset price movements at the time of the announcement. The one residual qualification, discussed below, is the temporal bias of the reader model.

The second contribution is empirical. We produce a narrative-based monetary policy surprise series for the ECB covering the full 1999–2025 period. The series exhibits three properties consistent with a well-behaved measure: a near-zero mean consistent with broadly informative communications, a compression of volatility during the Draghi presidency that reflects a combination of forward guidance and policy rates pinned near the effective lower bound, and the largest positive surprises concentrated in the 2022–2023 hiking cycle where communications systematically understated the pace of tightening. The September 2019 case — a rate cut classified as a hawkish surprise because the ECB delivered less easing than its own communications had signalled — illustrates the conceptual distinction between our measure and market-based alternatives: we capture the gap between what the ECB said and what it did, not what markets had priced in.

The third contribution is econometric. We provide evidence on the asymmetric transmission of ECB monetary policy surprises using Local Projections. The finding that contractionary surprises are substantially more powerful than expansionary ones adds to a growing literature on asymmetric monetary transmission (Cover, 1992; Tenreyro and Thwaites, 2016), extending direction-dependent asymmetry evidence to the ECB context using a narrative identification strategy that, by construction, does not incorporate the announcement-window asset-price response through which the information puzzle operates (Nakamura and Steinsson, 2018).

6.2 Limitations

Three limitations deserve explicit acknowledgement.

The structural break at January 2015 is the most consequential. Prior to that date, Agent IM reads Introductory Statements of Press Conferences; after it, the Accounts of the Governing Council — a substantially richer document that captures the internal deliberations of the Council rather than its public face. The F-test confirms a statistically significant break in the mean of the surprise series, and the two sources of the break — the document substitution and the monetary regime shift to the zero lower bound — cannot be separately identified. We note that a restricted post-2015-only estimation is too underpowered to be informative: with only $\sim$87 post-2015 meetings entering the monthly series, the effective observations at each LP horizon fall to fewer than 40, producing large standard errors and unreliable point estimates. We therefore treat the post-2015 period as a robustness context rather than a primary estimation window, and the full-sample analysis should be interpreted with the structural break caveat in mind.

The second limitation concerns the *a priori* nature of the aggregation weights. The weights combining the policy signal and the economic signal (equation (3.4)), the internal weights of each signal, and the weights of the economic conditions

index cannot be estimated from the data without introducing circular inference. The sensitivity analysis in Table 3.3 confirms that the results are stable across a range of alternative weight configurations, but the fundamental indeterminacy of the optimal weights remains an open methodological question.

The third limitation is the temporal bias inherent in any LLM applied to historical documents, which operates through two channels. The first is interpretive: the model may apply macroeconomic frameworks shaped by hindsight. The second, more serious, is outcome leakage: because Gemini 2.5 Flash was trained on data that likely describes the outcomes of past ECB decisions, the model may have memorised actual meeting results, contaminating its prior distributions and violating the ex-ante construction principle. Temperature and prompt constraints reduce the interpretive channel but cannot prevent memorisation. This remains a genuine open limitation that cannot be fully resolved without a model trained on pre-1999 text.

6.3 Directions for Future Research

The most immediate and natural extension is the construction of a market-based complement to the narrative surprise measure. Using options on Euribor futures — specifically caplets priced before and after each Governing Council meeting — it is possible to extract the rate distribution implied by financial markets immediately before each announcement. The Black (1976) caplet framework is the standard tool for this extraction, and historical data are available from the ECB Working Paper dataset of Puigvert-Gutiérrez and de Vincent-Humphreys (2010) for the period 1999–2010, with Bloomberg or Refinitiv providing coverage for the subsequent period. Comparing the narrative and market-based measures would allow direct identification of the component of ECB communications already pre-priced through informal channels — speeches, press interactions, market contacts — versus the component that constituted genuine institutional news. The decomposition is directly relevant to the critique of market neutrality raised in Chapter 3: if the two measures correlate strongly, it would suggest that official communications are the primary driver of market expectations; if they diverge systematically, it would indicate that informal information channels play a larger role than the official record reflects.

Two further extensions follow naturally from the structure of the dataset. The first is the analysis of heterogeneous transmission across core and peripheral euro area countries using panel Local Projections on country-level inflation and output data — a direct test of whether the asymmetry documented in Chapter 5 is amplified in economies with less developed capital markets and higher sovereign risk premia. The second is a formal analysis of communication transparency across pres-

idential eras, using the time series of posterior entropies from Agent II as a measure of guidance credibility. Preliminary evidence in Chapter 4 suggests that entropy is systematically lower during the 2013–2018 forward guidance era under Draghi; a more rigorous treatment would quantify the relationship between entropy reduction and surprise magnitude, testing whether more predictable communications translate into smaller transmission effects.

6.4 Closing Remarks

The transformation of central bank communication over the past three decades has made monetary policy more predictable, more accountable, and in many ways more effective. But predictability is not the same as certainty, and the gap between what a central bank communicates and what it decides remains a meaningful source of variation — one that, as this thesis documents, is associated with subsequent inflation dynamics in ways that are both statistically robust and economically substantial.

The finding that this gap matters more when it points in a restrictive direction than when it points in an expansionary one adds to a growing body of evidence that monetary policy transmission is not symmetric. Whether this asymmetry reflects nominal rigidities, financial frictions, or the particular conditions of the euro area is a question that this thesis opens rather than closes. What it suggests is that the ECB's official communications are an informative, quantifiable, and econometrically useful signal — one that appears to capture dimensions of monetary policy that market prices alone do not.

A

Additional Tables and Figures

The robustness tables supporting the pipeline validation in Chapter 3 are presented inline in the Chapter 3 appendix (Section 3.6) and are reproduced at their original locations: Table 3.2 (correlation structure of Agent IB sub-scores), Table 3.3 (sensitivity analysis on aggregation weights), and Table 3.4 (regime-interaction test).

B

Agent Prompt Templates

This appendix reproduces the system prompts supplied to the two LLM agents of the pipeline, Agents IM and IB. Both prompts use Google Gemini 2.5 Flash at temperature 0.1. The prompts are designed to enforce two methodological constraints: strict temporal grounding (the agent is anchored to the document publication date and must not use information released after that date) and structured output (the agent must return valid JSON matching the specified schema, with no free-text preamble or markdown formatting). Agents II and III perform no LLM call and therefore have no prompt; their deterministic computations are documented in the final two sections below for completeness.

B.1 Agent IM — Monetary Intelligence

Agent IM processes the most recent ECB Account (post-2015) or Introductory Statement of the Press Conference (pre-2015) and extracts the Governing Council's policy stance.

```
You are Agent IM (Monetary Intelligence), a specialist in ECB monetary
policy. Your task: analyse the provided ECB Account/Press Conference from
meeting t-1 and produce a structured JSON assessment of the Governing
Council's monetary stance.

CRITICAL RULES:
- Temporal anchor: you only have information up to the ACCOUNT PUBLICATION
  DATE provided. Never infer information from data released after that date.
- All probability fields must sum to exactly 1.0.
- Return ONLY valid JSON - no markdown, no preamble, no explanation outside
  JSON.

OUTPUT JSON SCHEMA (all fields required):
{
```

```
  "meeting_analyzed": "YYYY-MM-DD",
  "account_pub_date": "YYYY-MM-DD",
  "target_meeting": "YYYY-MM-DD",
  "hawkishness_score": float [-1.0 very dovish to +1.0 very hawkish],
  "hawkishness_confidence": float [0.0-1.0],
  "hawkishness_rationale": "concise rationale (max 150 words)",
  "council_division": {
    "hawks_fraction": float,
    "doves_fraction": float,
    "centrists_fraction": float,
    "dissent_detected": bool,
    "dissent_description": "description or None detected"
  },
  "forward_guidance": {
    "type": "Odyssean|Delphic|None",
    "verbatim_extract": "exact quote or None",
    "implied_direction": "hike|hold|cut|ambiguous"
  },
  "uncertainty_score": float [0=certain to 1=maximum uncertainty],
  "baseline_distribution": {
    "hike_large": float,
    "hike_standard": float,
    "hold": float,
    "cut_standard": float,
    "cut_large": float
  },
  "shock_indicators": ["up to 3 phrases signalling potential surprise"],
  "nonstandard_tools": ["APP", "PEPP", "TPI", etc. if mentioned],
  "key_phrases": ["up to 5 most policy-relevant phrases verbatim"]
}
```

B.2 Agent IB — Economic Conditions

Agent IB processes the most recent Economic Bulletin (or Monthly Bulletin pre-2015) and qualifying speeches, extracting five economic condition scores weighted according to the ECB's price stability mandate.

```
You are Agent IB (Economic Conditions), a specialist in ECB economic
analysis. Your task: read the provided ECB Economic Bulletin / Monthly
Bulletin excerpt and extract quantitative economic condition scores that the
Governing Council would use for rate decisions.
```

```
ECB MANDATE PRIORITY:
- Price stability (HICP inflation) is the PRIMARY mandate.
- Score -1.0 = extremely dovish/weak conditions (far below target/trend)
- Score +1.0 = extremely hawkish/strong conditions (far above target/trend)
- Score  0.0 = neutral / on target

CRITICAL RULES:
- Only use information from the document text provided.
- Temporal anchor: you only know what was available as of the BULLETIN
  PUBLICATION DATE.
- Return ONLY valid JSON - no markdown, no explanation outside JSON.

OUTPUT JSON SCHEMA:
{
  "inflation_outlook":   float [-1.0 to +1.0],
  "growth_prospects":    float [-1.0 to +1.0],
  "labor_market":        float [-1.0 to +1.0],
  "policy_stance":       float [-1.0 to +1.0],
  "inflation_rationale": "key HICP/inflation evidence cited (max 100 words)",
  "growth_rationale":    "key GDP/activity evidence cited (max 80 words)",
  "labor_rationale":     "key employment evidence cited (max 80 words)",
  "policy_rationale":    "explicit rate/tool signals mentioned (max 80 words)",
  "confidence": float [0.0-1.0]
}

[Speech variant: same schema with additional field]
  "key_signals": ["up to 3 key phrases"]
```

B.3 Agent II — Synthesis

Agent II combines the outputs of Agents IM and IB through a deterministic softmax update rather than a separate LLM call. The combined signal is computed as:

$$
\begin{aligned}
s_{\text{policy}} &= 0.60\, h_{\text{IM}} + 0.20\, s_{\text{FG}} + 0.20\, e_4 \\
s_{\text{econ}} &= 0.70\, s_{\text{IB}} + 0.30\, e_{\text{speech}} \\
\tilde{s} &= 0.55\, s_{\text{policy}} + 0.45\, s_{\text{econ}}
\end{aligned}
$$

where $s_{\text{FG}} \in \{+0.6, 0.0, -0.6\}$ maps the forward guidance implied direction (hike / hold or ambiguous / cut), e_4 is the bulletin's implied policy stance, and $s_{\text{IB}} = 0.50\, e_1 + 0.30\, e_2 + 0.20\, e_3$ is the aggregate economic index over the three macroeco-

nomic conditions scores (inflation, growth, labour). The combined signal $\tilde{s}$ updates the Agent IM prior distribution via:

$$\log \mathbf{p}^{\text{post}} = \log \mathbf{p}^{\text{prior}} + \tilde{s} \cdot [+2, +1, 0, -1, -2]$$

followed by a softmax normalisation. No LLM call is made at this stage; the update is fully deterministic.

B.4 Agent III — Surprise Calculation

Agent III computes three surprise measures from the posterior distribution produced by Agent II and the actual ECB rate decision. No LLM call is made; the calculation is fully deterministic.

$$\begin{aligned} s_t^{\text{mech}} &= \Delta_{i,t}^{\text{actual}} - \mathbb{E}[\Delta_{i,t}] \\ s_t^{\text{sal}} &= -\log(p_{k^*}^*) \\ s_t^{\text{norm}} &= s_t^{\text{mech}} \,/\, \hat{\sigma}_{t-1} \end{aligned}$$

where $\mathbb{E}[\Delta_{i,t}] = \sum_k p_k^* \Delta_{i,k}$ is the probability-weighted expected rate change, k^* is the bin corresponding to the actual decision, $p_{k^*}^*$ is its posterior probability, and $\hat{\sigma}_{t-1}$ is the rolling standard deviation of the preceding 12 meetings. The direction classification uses thresholds of ± 5 bp: $s_t^{\text{mech}} > 5$ bp is labelled *hawkish surprise*; $s_t^{\text{mech}} < -5$ bp is labelled *dovish surprise*; all others are *no surprise*.

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